

When does framing shape sustainable choice?

An online randomized experiment on gain/loss framing, instruction-based time pressure, environmental attitude, and repeated consumer decisions.

Research question

Does loss framing increase sustainable product choice, and does that pattern change under time pressure or across participants with different environmental attitudes? The thesis also explores decision consistency as a possible boundary condition.

Study design

Design	2 × 2 between-participant experiment: gain/loss framing × instruction-based time pressure
Task	Repeated binary sustainable vs. conventional product choices; up to 24 trials per participant
Analysis sample	101 participants; 2,424 constructed trial rows; 2,107 valid choice observations; 2,197 valid reaction-time observations
Methods	Decision-level logistic regression with participant-clustered standard errors; log reaction-time regression; exploratory entropy measure from repeated choices

From survey export to analysis

The raw online-experiment export was participant-wide, while the research question required trial-level analysis. The saved notebook restructures repeated choices into decision records, flags quality indicators, joins participant-level attitude measures, and fits models that account for repeated observations within each participant.

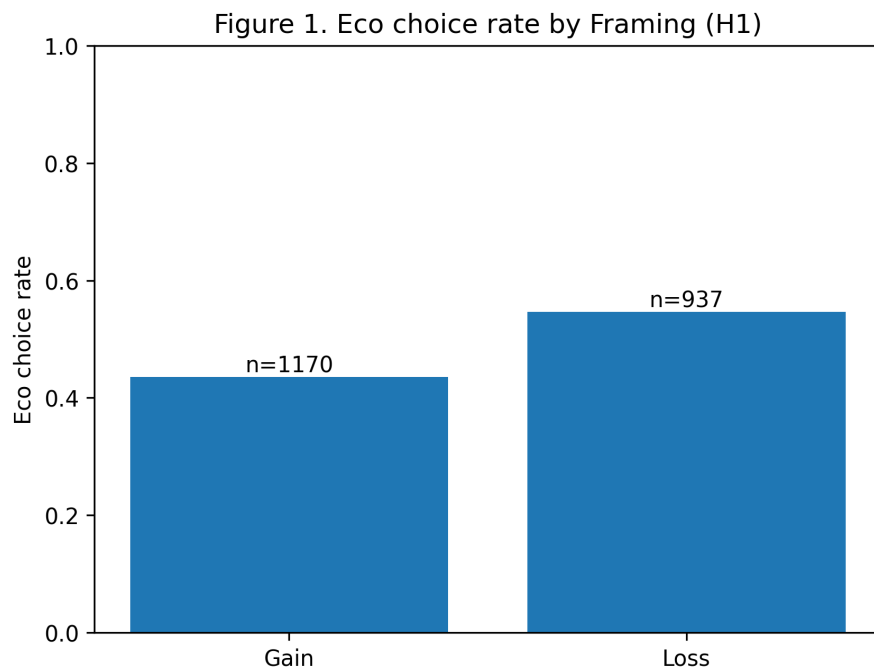
Workflow evidence

01	Preserve the raw export and inspect its columns and embedded labels.
02	Reshape repeated trial answers and timing fields from wide to long.
03	Create trial-level choice and reaction-time variables; retain diagnostic flags.
04	Attach participant-level environmental attitude and derive repeated-choice entropy for exploratory analysis.
05	Fit choice and reaction-time models; cluster inference by participant.
06	Compare model estimates, saved tables, and descriptive plots before interpreting results.

The reported choice models use 2,107 valid observations, not all 2,424 reshaped rows. The reaction-time models use 2,197 valid timing observations. This distinction matters because 317 constructed trial rows do not contain a valid binary choice.

What the evidence shows

The estimated loss-framing effect is positive, but imprecise. The primary model reports OR 1.56, 95% CI [0.82, 2.97], $p = 0.177$. The data do not establish a reliable average framing effect in this sample.



Descriptive eco-choice rates among valid choice observations; the regression uses participant-clustered standard errors.

Other results

In the model with gain framing as the reference condition, higher pre-task environmental attitude is associated with greater odds of sustainable choice (OR 3.51, 95% CI [2.14, 5.78], $p < 0.001$). This is an adjusted association, not a randomized treatment effect.

The instruction-based time-pressure condition does not show a clear reaction-time difference in the saved final model (coefficient 0.036 on log RT, 95% CI [-0.175, 0.246], $p = 0.738$).

What this demonstrates

Research design

Translated a consumer-behavior question into a randomized 2×2 design with repeated choice observations.

Data preparation

Restructured participant-wide survey output into trial-level records and joined participant measures to decisions.

Statistical analysis

Used logistic and reaction-time models, clustered uncertainty at the participant level, and distinguished descriptive patterns from inferential results.

Quality judgment

Kept the weak average framing estimate and failed time-pressure check visible; treated entropy-based subgroup analysis as exploratory.

Evidence boundary

The experiment has 101 participants from an online bilingual task. The time-pressure manipulation was instructional rather than enforced. Entropy is derived from observed choices and cannot be interpreted as a pre-treatment causal trait. The source notebook's attention-check flag logic needs revalidation, so attention-based robustness claims are not used in this case summary.

Evidence and files

Full thesis

The complete 58-page thesis is included as a public copy. The identifying student number and email on the original cover were removed; the thesis body is retained.

Analysis evidence

A short selection of the original notebook's data restructuring and model code is included. The raw participant-level survey export is not included.

Saved outputs

Aggregate model tables and the descriptive framing figure are provided so readers can inspect the estimates used in this summary. The notebook itself contains historical code and execution paths; the excerpts are evidence, not a clean one-click reproduction package.

Research record note

The submitted manuscript contains a sample-size statement that does not match its saved regression output. This case uses 2,107 for valid choice-model observations and 2,197 for valid reaction-time observations, as recorded in the cleaned dataset and model outputs.

Thesis and case materials prepared for portfolio display. Findings are limited to this sample and design.